Exp: 10

29/9/26

Implementation of Subnetting In CISCO Packet Tracer Simulation

Aim:

To implement classless IP subnetting using CISCO Packet Tracer and verify connectivity between different subnets.

Procedure:

1) Open Cisco Packet Tracer and create a new network topology.

2) Add routers, switches and PC's to the workspaces.

3) Connect the devices using appropriate cables

4) Divide the network 192.168.1.0/24 into smaller subnets using the /27 subnet mask.

5) Configure the IP addresses and subnet mask for the router interfaces

6) Configure the PC's with suitable IP addresses, subnet masks and default gateways.

7) Configure the switch ports and router interfaces

8) Use the PING command to test connectivity between the devices

9) Verify that the packets are successfully transmitted between the connected networks.

Advantages:

* Efficient use of IP address
* Reduces network congestion
* Improves network management

IP Addressing:-

Network	Subnet mask	Router IP	PC IP Range
192.168.1.0 /27	255.255.255 .224	192.168.1.1	192.168.1.11-15
192.168.2.0 /27	255.255.255 .224	192.168.2.1	192.168.2.11-15
192.168.3.0 /27	255.255.255 .224	192.168.3.1	192.168.3.11-15
192.168.4.0 /27	255.255.255 .224	192.168.4.1	192.168.4.11 -15

Result:-

The classless IP subnetting was successfully implemented using CISCO Packet Tracer.